

Chapter - 8

3 hrs.

7/8 marks.

Data Link Control & protocol.

The main responsibility of the data link layer are framing, flow control, error control and software implemented protocols that provided smooth & reliable transmission betⁿ nodes.

Framing :

Framing deals with how to organize the bits that are carried by the physical layer. Frame can be of fixed or variable size.

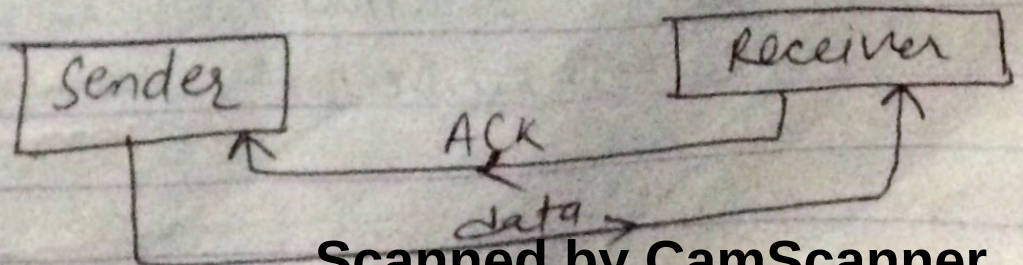
- * Fixed size framing : There is no need for defining the boundaries of the frames because the size itself can be used as a delimiter. eg (ATM cell)
(~~cell~~ fix)
- * Variable size framing : The size of frames are variable it may be use variable no. of character or variable no. of bits depending on which two protocols are used.

Flow control :

- Flow control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement from the receiver.
- Any receiving device has a limited speed at which it can process incoming data & a limited amount of memory in which to store incoming data.
- Thus the flow of data must not be allowed to overwhelm (आवरण) the receiver.
- The protocols used for flow control are.

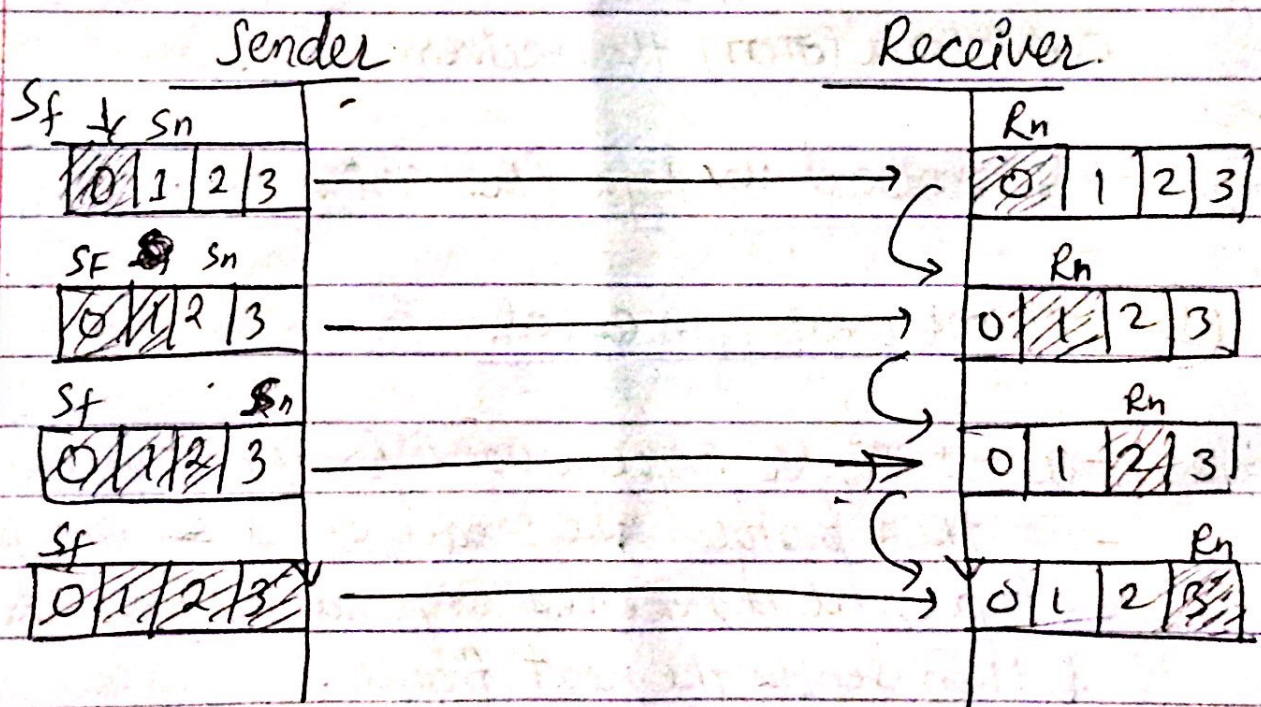
① Stop & wait flow control.

- It is a simple protocol used for flow control.
- In this protocol the sender sends ~~the~~ one frame, stop until it receives confirmation from the receiver & then sends the next frame.
- Here for data frames the communication is unidirectional. but auxiliary Ack frames travel from other direction.



② Sliding Window.

- This protocol allows multiple frames to be in transit.
- Receiver sends ack with sequence no. of anticipated frame
- Sender ~~may~~ maintains list of sequence no. it can send & receiver maintains list of sequence no. it can receive



S_f → Send First

S_n → Send Next

R_n → Receive next

- Piggybacking is a feature for bidirectional flow control using sliding window protocol.
- If two stations exchange data, each needs to maintain two windows, one for transmit & one for receive & each side needs to send the data & ack to the other.

Error Control

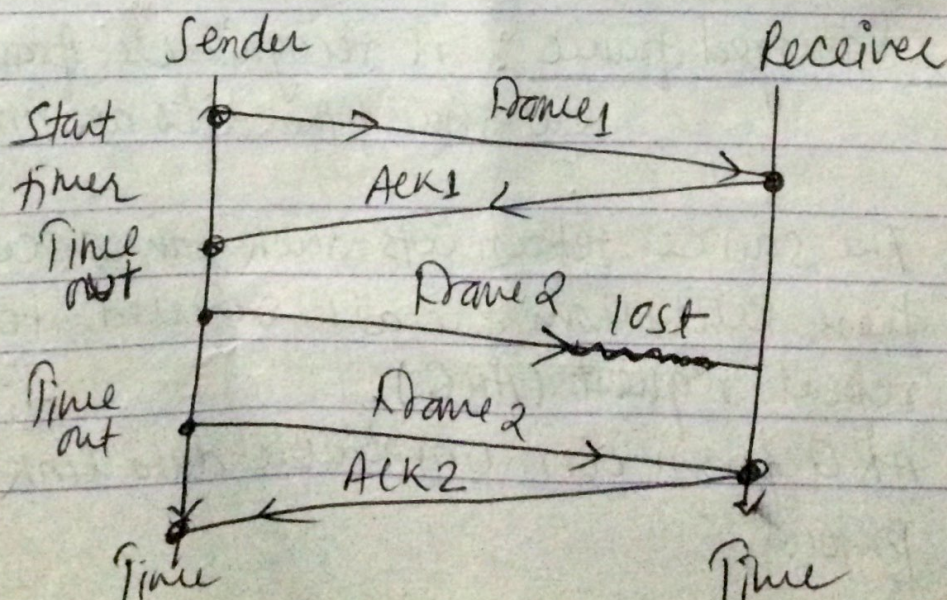
- Error control is both error detection & error correction.
- If any frames lost or damaged in transmission then error control protocol allows the receiver to inform & retransmits those frames.
- In the transmission of sequence of frames there is possibility of two types of error.
- Lost frame : A frame that fails to arrive at the other side.
- Damaged frame : A recognizable frame does arrive but some of the bits are in error.
- The source retransmits such frames collectively these mechanism are all referred to as automatic repeat request (ARQ).
- ARQ turns an unreliable data link error control protocol.

- ① stop & wait ARQ.
- ② Go-Back-N ARQ.
- ③ Selective-reject ARQ.

1 Stop & wait ARQ.

- It is based on stop & wait flow control technique the source transmits a single frame & then must wait on acknowledgement (ACK). Here two sorts of errors could occurs. i.e

- ① The frame that arrives at the destination could be damaged or lost hence to account these possibility the source station is equipped with a timer. After the frame is transmitted the same station waits for an acknowledgement. If no ack. is received by the time, the time expires, then the frame is sent again.
- ② The ack of the frame may be damaged or lost.



2 Go-Back-N ARQ.

- This form of error control is based on sliding window flow control.
- In go-back-N ARQ, a station may send a sequentially numbered series of frames.
- The number of unacknowledged frames outstanding is determined by window size, using the sliding window flow control technique.
- While no error occurs the destination will acknowledge (RR - Receive Ready) incoming frames as usual.
- If the destination detects an error in a frame it sends a -ve ack i.e. (REJ - Reject) for that frame.
- Until rejected frame not received correctly it does not receive other frames.
- So transmitter transmits that rejected frame as well as other frames.
- The receiver receives 1st rejected frame again then other transmitted.

3 Selective - Reject ARQ.

- It is similar to go-back-N ARQ.
- It is also based on sliding window flow control.
- Here if the destination station detects an error in a frame, it sends a -ve ack for that frame.

- The destination station will discard that frame & retransmit only that frame rather than in go-back-N ARQ. Where along with that error frame & succeeding frames.

Important Q.

- ① Describes flow control & error control. How does Go-Back-N ARQ works? 2013
- ② What is ARQ? Explain different error control technique 2006.
- ③ What is ARQ How does Stop & Wait ARQ & sliding window ARQ deals with error? 2011
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